

INTRODUCTION TO STATISTICAL METHODS

Correlation and Regression

Pearson correlation, the least-squares line, slope and intercept, and r-squared.

Companion reader · Read beside the slides

How to use this companion. Colour boxes guide you: **blue** intuition, **amber** everyday picture, **green** worked example, **red** watch out, **purple** key takeaway. Read it next to the slides, in order.

1 Correlation: strength of a linear relationship

The **Pearson correlation** r measures how tightly two variables move together in a straight line. It ranges from -1 (perfect negative) through 0 (no linear relationship) to $+1$ (perfect positive):

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}.$$

The sign gives the direction; the magnitude gives the strength.

⚠ Watch out

Correlation is not causation, and r only captures *linear* association — a perfect U-shape can have $r = 0$. Always look at the scatter plot before trusting a correlation.

2 Regression: the best-fit line

Linear regression finds the line $y = a + bx$ that minimises the sum of squared vertical errors (**least squares**). The slope and intercept are

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}, \quad a = \bar{y} - b\bar{x}.$$

📎 Worked example: correlation and regression from data

Data: $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 5)$. Means $\bar{x} = 3$, $\bar{y} = 4$.

$$\sum(x - \bar{x})(y - \bar{y}) = (-2)(-2) + (-1)(0) + 0(1) + 1(0) + 2(1) = 6, \quad \sum(x - \bar{x})^2 = 10, \quad \sum(y - \bar{y})^2 = 6.$$

Slope $b = \frac{6}{10} = \mathbf{0.6}$; intercept $a = 4 - 0.6(3) = \mathbf{2.2}$, so the line is $\hat{y} = 2.2 + 0.6x$. Correlation

$$r = \frac{6}{\sqrt{10 \cdot 6}} = \frac{6}{7.746} = \mathbf{0.775}, \text{ and } r^2 = \mathbf{0.60}: 60\% \text{ of the variation in } y \text{ is explained by } x.$$

3 Interpreting the fit

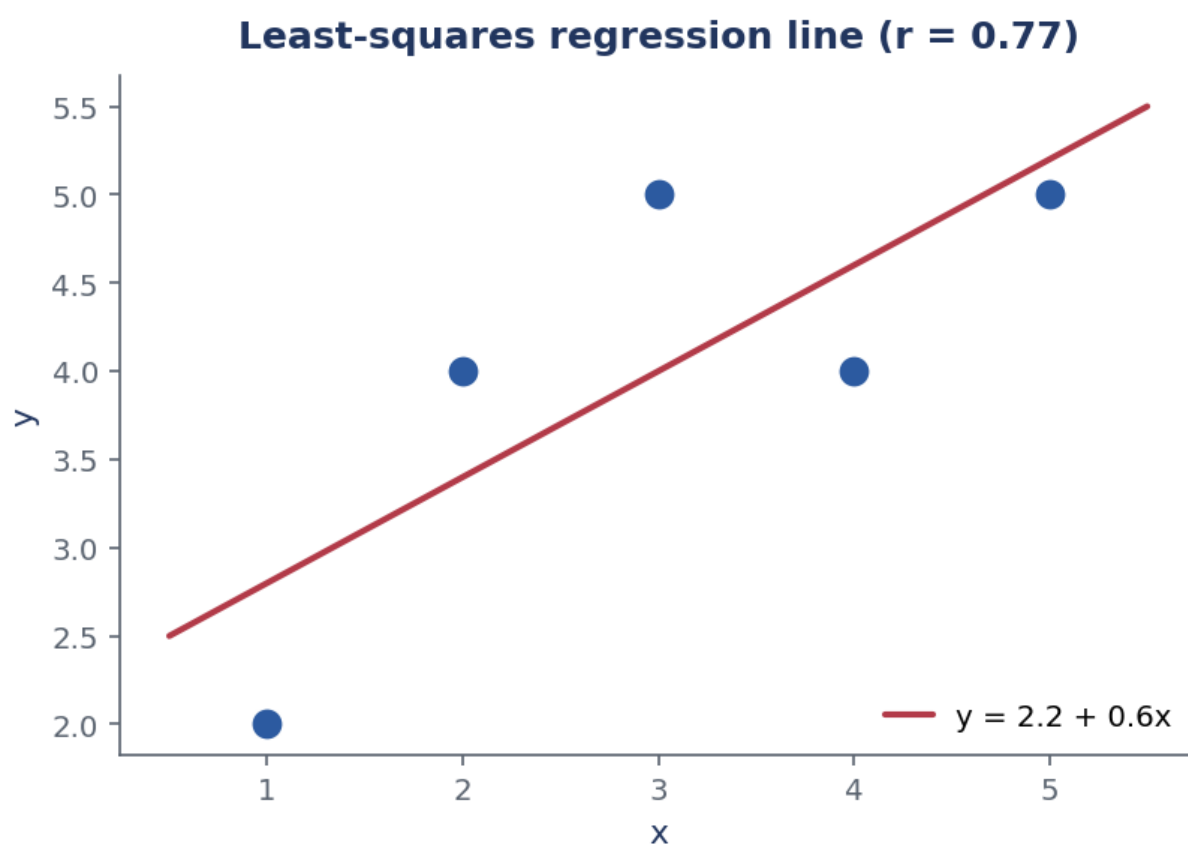


Figure 1: The least-squares line $\hat{y} = 2.2 + 0.6x$ is the straight line that minimises the total squared vertical distance to the points.

r^2 (the **coefficient of determination**) is the fraction of y 's variance explained by the model — here 0.60. The slope b says how much $\hat{y}$ changes per unit of x (here +0.6). Use the line to *predict* $\hat{y}$ at a new x — but only within the observed range; **extrapolating** far outside it is unreliable.

✓ Key takeaway

Correlation $r \in [-1, 1]$ measures the strength and direction of a *linear* relationship (but not causation). **Regression** fits the least-squares line $y = a + bx$ with $b = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}$ and $a = \bar{y} - b\bar{x}$ (worked: $\hat{y} = 2.2 + 0.6x$, $r = 0.775$, $r^2 = 0.60$). r^2 is the fraction of variance explained.